

Kathmandu University
Department of Computer Science and Engineering
Dhulikhel, Kavre



A Project Report
on
“altEco”

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Submitted by
Rochak Adhikari (04)
Suyog Basnet (08)
Aditya Pokharel (32)
Nikunj Pokharel(33)
Nerish Shrestha (53)

Submitted to
Mr. Suman Shrestha
Department of Computer Science and Engineering

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Bona fide Certificate

This project work on

“altEco”

is the bona fide work of

“Rochak Adhikari”

“Suyog Basnet”

“Nikunj Pokharel”

“Aditya Pokharel”

“Nerish Shrestha”

who carried out the project work under my supervision.

Project Supervisor

Dr. Rajani Chulyadyo

Assistant Professor

Department of Computer Science and Engineering

Date: 2025/05/08

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This journey has been an invaluable learning experience, helping us grow both as developers and as a team. We sincerely appreciate the support, resources, and guidance that have contributed to making this project a success.

Abstract

With the increasing global emphasis on environmental sustainability, consumers are seeking reliable ways to make eco-conscious purchasing decisions. However, challenges such as greenwashing, inconsistent labeling, and limited access to trustworthy environmental data hinder the identification of genuinely sustainable products. This project provides a dedicated platform that leverages machine learning to recommend eco-friendly consumer products based on user preferences and standardized sustainability metrics. The system analyzes diverse data points—including material sourcing, manufacturing practices, and eco-certifications—to generate personalized recommendations and assign sustainability scores to products. Additional features include educational content, impact tracking, and community-driven insights to promote informed and responsible consumption. By offering accessible, actionable information and alternatives, the platform aims to empower users to reduce their environmental footprint, foster behavioral change toward sustainable lifestyles, and support the broader goals of waste reduction, resource conservation, and decreased carbon emissions.

Keywords: *Sustainability, Machine Learning, Greenwashing, Carbon footprint, Data Analysis*

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Acronyms/Abbreviations

- AI: Artificial Intelligence
- API: Application Programming Interface
- CSS: Cascading Style Sheets (as in Tailwind CSS)
- DB: Database (implied in context, e.g., MongoDB)
- GPU: Graphics Processing Unit
- IDE: Integrated Development Environment
- JS: JavaScript (as in ReactJS, Node.js, Express.js)
- ML: Machine Learning
- OS: Operating System (as in Windows, macOS, Linux)
- RAM: Random Access Memory
- SSD: Solid State Drive
- UI: User Interface

Chapter 1 Introduction

1.1 Background

With growing awareness of the environmental impact of consumer habits, individuals are increasingly motivated to make eco-conscious decisions in their daily purchases. (Ashley Reichheld, 2023) The push for sustainability has led to a rising demand for products that are ethically sourced, biodegradable, recyclable, or have a low carbon footprint. However, despite this demand, consumers often face difficulty identifying genuinely sustainable products due to greenwashing, inconsistent labeling, and limited access to reliable environmental data. (Şenyapar, 2024)

Most current e-commerce platforms and product search tools are not equipped to provide detailed insights into the sustainability of the products they feature. While some platforms highlight eco-tags or offer generic filters like “organic” or “vegan,” these features lack depth and consistency. As a result, consumers are left without a comprehensive way to compare the environmental impact of their choices or to discover more responsible alternatives that align with their values. (Emmanuel Mireku, 2024)

This has now changed with the introduction of a dedicated platform leveraging artificial intelligence to empower sustainable shopping. By analyzing a wide range of data points such as material sourcing, manufacturing practices, and eco-certifications the system generates personalized recommendations for eco-friendly products. The platform also offers educational content, impact tracking features, and community-driven insights, helping users become more informed and proactive in reducing their environmental footprint. Through this approach, the solution aims to empower everyday consumers to contribute meaningfully to sustainability, not just through awareness but through measurable actions.

1.2 Objectives

The major objectives of altEco are listed below:

- i. To recommend eco-friendly consumer products based on user preferences and sustainability metrics.
- ii. To assign and display sustainability scores for products using standardized environmental criteria.
- iii. To allow customization of recommendations based on ethical and ecological filters.
- iv. To track users' environmental impact through eco-friendly product choices.

1.3 Motivation and Significance

In today's world, environmental sustainability has become a global priority, yet many individuals remain unaware of the harmful impacts of everyday products they use. (Sousa, 2023) The project altEco is driven by the need to provide accessible and practical information that helps users identify commonly used objects and discover their eco-friendly alternatives. By offering clear, sustainable options, the project encourages more responsible consumption habits and supports a transition toward greener lifestyles. This initiative is significant not only in promoting individual awareness but also in contributing to broader environmental goals such as waste reduction, resource conservation, and decreased carbon footprint.

Chapter 2 Related Works

Some existing platforms related to our project are mentioned below:

2.1 Yuka

Yuka (Yuka, 2025) is a mobile app that helps users make healthier and safer choices by evaluating food and cosmetic products. It uses a rule-based scoring system to assess nutritional content, additives, and the risk level of ingredients, assigning products a score from 0 to 100 and provides recommendations for healthier or safer alternatives. Yuka bases their score based on Nutri-Score. The Nutri-Score is calculated by assigning points based on the nutritional content per 100 grams or milliliters of the product. It has two main components:

1. Negative points (N): These are assigned for ingredients that should be limited in the diet, including energy (calories), sugars, saturated fats, and sodium (salt). Each of these can receive between 0 to 10 points based on their amounts.
2. Positive points (P): These are assigned for beneficial components such as the percentage of fruits, vegetables, legumes, nuts, fiber content, and protein. These positive points typically range from 0 to 5.

The overall Nutri-Score value is calculated by subtracting the total positive points from the total negative points ($N - P$). This value can range roughly from -15 (for the healthiest products) to +40 (for the least healthy). Based on the score, the product is classified into one of five letter grades: A (best) to E (worst).

2.2 Green Choice

Green Choice (Green Choice, 2025) is a mobile application designed to assist users with their food choices. They not only take into consideration the nutritional value of various options but also how eco-friendly they are. They base their points on a 1-100 scale using the nutritional value of the nutrients included in the product. Specifically, they compare the nutritional value of each nutrient with the Daily Value as approved by the FDA. It awards points if the product contains 5% or more

of the DV of a beneficial nutrient per serving, and subtract points when a product contains 10% or more of the DV of a discouraged nutrient.

Positive Points

The following are the beneficial nutrients taken into consideration by the GS-Nutrition Score.

- Protein
- Fiber
- Potassium
- Calcium
- Vitamin D
- Iron
- Omega-3s

Negative Points

The following are the discouraged nutrients taken into consideration by the GS-Nutrition Score:

- Saturated Fat
- Cholesterol
- Sodium
- Added Sugar

Chapter 3 Requirement Analysis and System Design

This chapter details altEco's systematic development approach, from comprehensive requirement analysis and competitive research through wireframe design, data planning, and database architecture creation for sustainable product recommendations.

3.1 Requirement Analysis and Gathering

The development of altEco began with comprehensive requirement analysis to understand the needs of environmentally conscious consumers and the challenges they face in identifying sustainable products. We conducted:

3.1.1 Functional Requirements

The core functionality of the altEco platform revolves around allowing users to discover sustainable products based on their preferences. Below is a detailed breakdown of the main actions and features required to make the system function effectively.

User Actions on the Website:

- Search for eco-friendly products based on product name, brand, tags, or sustainability keywords.
- View sustainability scores and environmental impact details of listed products.
- Use the Eco Score Calculator to estimate a product's sustainability by inputting its attributes.
- Scan barcodes to find product data or alternatives from the database.
- Save products to collections such as Favorites, Review Later, or Not Interested.
- Register and log in to create a personalized profile and track their environmental impact.
- Set and update eco-preferences (e.g., plastic-free, plant-based) to tailor recommendations. Access educational content, articles, and FAQs on sustainable consumption.

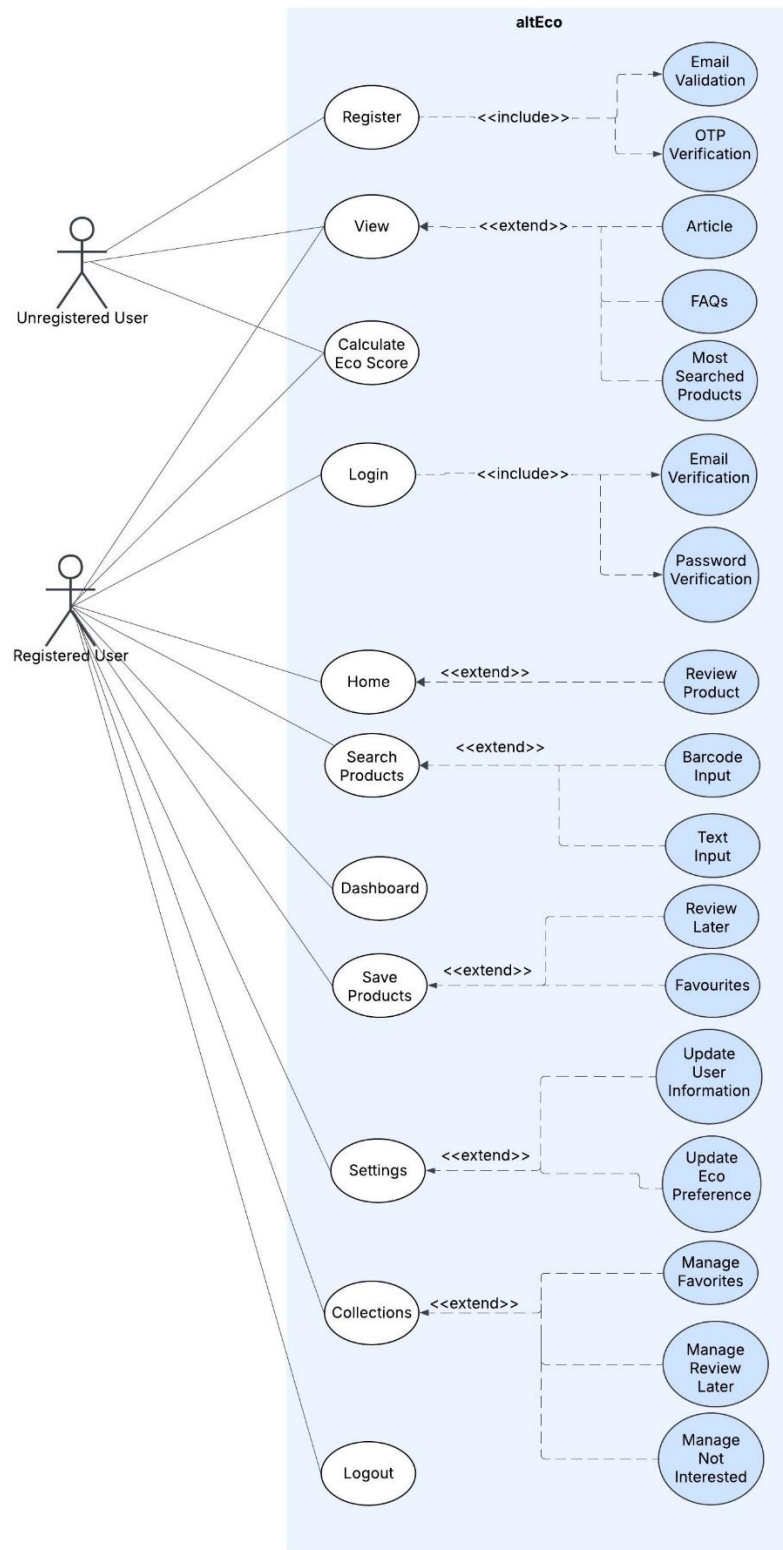


Figure 3.1.: Usecase diagram

3.2 Wireframe Design and Page Flow

We began by sketching wireframes in our notes to plan the layout and flow of the system. These wireframes included pages like the home page, login and registration, product search, product detail, set preferences, eco score calculator, review later page, and the user dashboard. From these sketches, we decided how each page would connect. For example, after a successful login, the user would be directed to the home page. From there, they could go to the search product page, view product details, save items to review later, set their preferences, or use the eco score calculator. This helped us understand the full user journey through the system.

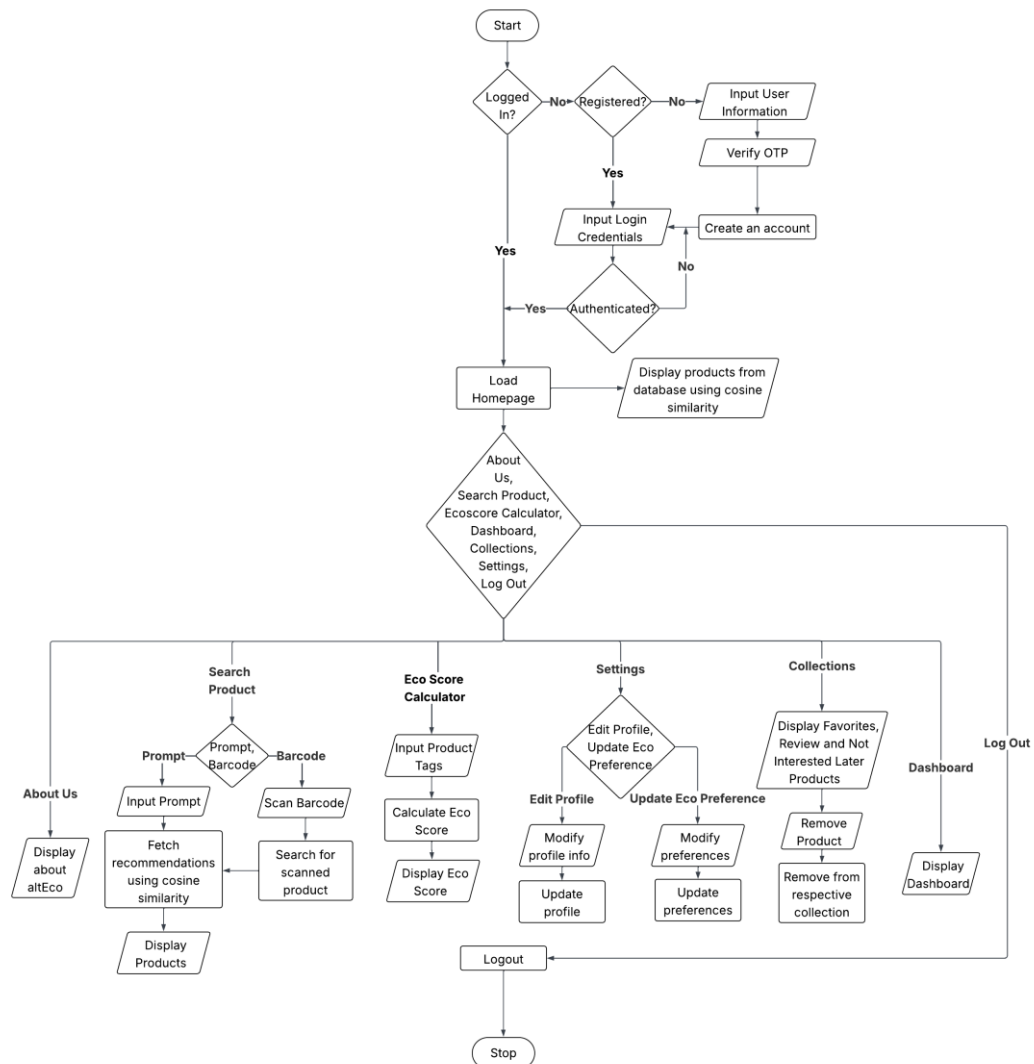


Figure 3.2.: Flowchart

3.3 Brainstorming Data Requirements

After deciding on the page flow, we discussed what data each page would need. Based on the wireframes, we thought about what types of information would be shown or stored. For example, the product detail page would need product information, the review later page would need to store selected items, and the preferences page would need user-specific settings. We listed out the types of data we would need across the system and talked about how they would be used. This step helped us get a basic idea of what kind of data our system had to handle.

3.4 Database Design

Once we had the data types figured out, we created an Entity-Relationship Diagram (ERD) to show how the different pieces of data relate to each other. After that, we made a schema diagram to define the tables, their fields, and how they connect. We made sure the schema was organized in a way that avoided unnecessary repetition and kept the data consistent. This design gave us a clear structure to follow when we later built the database during development.

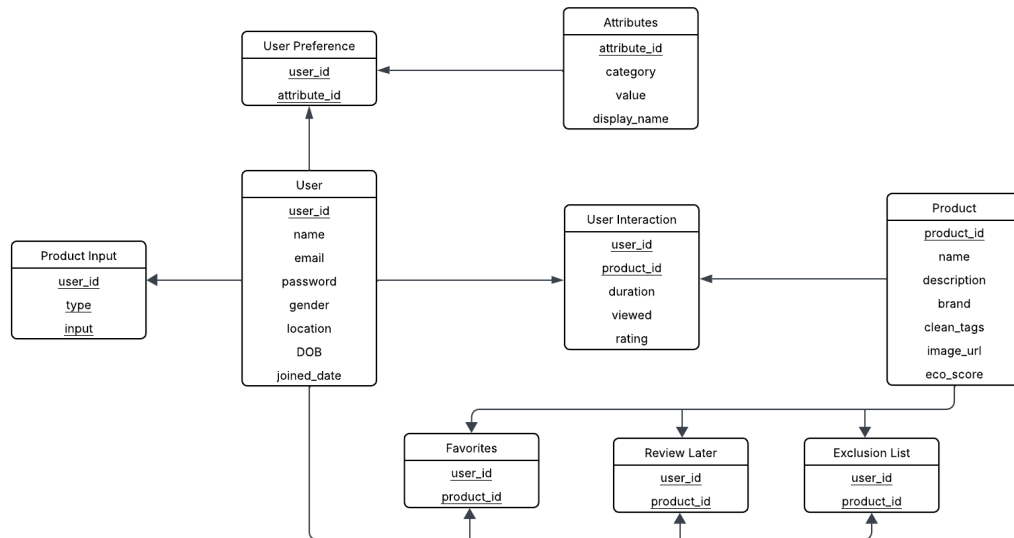


Figure 3.4.: Schema Diagram

Figure 3.4 illustrates the schema used for altEco. The core entities include Users with personal details and preferences and Products, each with specific descriptions, brands, and eco scores. Users interact with products through various lists (Favorites, Review Later, Exclusion List), and each interaction is logged with data such as

duration and user ratings. Products are further characterized by attributes, sustainability aspects, tags, and certifications, enabling nuanced analysis and recommendation features. The relationships link users to products and preferences, and products to attributes, tags, and certifications, supporting personalized and sustainable product discovery.

Chapter 4 Implementation and Achievement

This chapter describes the system setup, development process, and key features that were implemented during the creation of the altEco platform. It also highlights what was achieved throughout the project.

4.1 System Setup

The development of altEco required both software and hardware support on the developer and user sides.

Developer Requirements:

- Operating System: Windows 10/11, macOS, or any Linux distribution
- Frontend: ReactJS, Tailwind CSS, HTML, TypeScript
- Backend: Node.js, Express.js
- Database: PostgreSQL
- Machine Learning Tools: Python, Scikit-learn, Pandas, NumPy

User Requirements:

- Hardware: Any modern system with Intel Core i3 or better, 4 GB RAM, and integrated graphics
- Software: A recent version of Chrome, Firefox, Edge, or Safari on Windows, macOS, or Linux
- Connectivity: Stable internet connection (minimum 5 Mbps recommended)

4.2 Development Process

We began with designing Use Case Diagrams and Entity-Relationship (ER) Diagrams to plan the system's functionality and database structure. These helped us understand the key features and how users would interact with the platform

Frontend and backend development were carried out in parallel. The frontend focused on pages like the landing page, product recommendation section, and user authentication. The backend handled data processing, authentication, and communication with the database.

Basic machine learning logic was also integrated to improve the recommendation system. We used cosine similarity to compare product tags and suggest alternatives that are more sustainable. This approach made the recommendations feel more relevant to user preferences.

4.3 EcoScore Calculator

A key feature we implemented is the Eco Score Calculator, which measures how eco-friendly a product is. Products are described using tags like “organic”, “plastic”, or “plant-based”. Each tag has a weight: positive for good traits and negative for harmful ones.

To avoid counting the same idea more than once (e.g., “organic” and “bio”), similar tags are grouped together using a display map. The score is calculated using the formula:

$$\text{EcoScore} = 50 + \frac{100}{\pi} \arctan \left(\sum_{i=1}^{\infty} \text{weight}_i \right)$$

This formula gives a balanced score between 0 and 100, centered around 50. The calculator was built in the backend using TypeScript and Express.js and works by receiving product tags through a POST request and returning the calculated score.

4.4 Achievements

By the end of the project, we were able to deliver a working platform that supports eco-conscious decision-making in a simple and user-friendly way. Some of the main achievements include:

- A functioning recommendation system based on tag similarity
- A custom-built eco score calculator that helps users evaluate product impact
- A clean and responsive interface developed using modern frontend tools
- Smooth communication between the frontend, backend, and database
- A flexible system architecture that can be expanded with more features and data in the future

Although there were time constraints, we were able to complete the core features and test them successfully. The platform is now in a state where it can be used and improved further.

Chapter 5 Conclusion and Recommendation

The altEco project has successfully delivered a comprehensive web-based platform that addresses the critical need for accessible, reliable sustainability information in consumer decision-making. Through the integration of modern web technologies, robust database design, and user-centric features, we have created a solution that empowers environmentally conscious consumers to make informed purchasing decisions.

The project successfully implemented a sophisticated attribute-based recommendation system using PostgreSQL, enabling personalized product suggestions based on comprehensive user preferences. The platform features seven distinct preference categories—from animal ethics to additive awareness—providing users with granular control over their sustainability criteria. The responsive, accessible user interface ensures seamless experience across devices while maintaining the environmental theme through thoughtful design choices.

altEco addresses the growing consumer demand for transparency in product sustainability by providing clear, actionable information that was previously difficult to access or compare. The platform's eco-scoring system and personalized recommendations help users navigate the complex landscape of sustainable products while avoiding greenwashing tactics commonly employed in the marketplace.

5.1 Limitations

- **Simple ML Solution:** At present, the recommendation engine is built using cosine similarity, which, while effective for generating basic matches, does not yet leverage more advanced machine learning techniques that could capture deeper patterns in user preferences.
- **Limited Use of User Interaction Data:** Although the system records valuable interaction data such as product views, ratings, and time spent, these insights are not currently factored into the recommendation process, limiting the potential for highly personalized suggestions.

- **Insufficient Sustainability Certification Data:** The platform's eco-score calculations are based on available product attributes but are constrained by the absence of extensive sustainability certification data, which could otherwise strengthen the reliability and transparency of product assessments.
- **Lack of Two-Factor Authentication (2FA):** While the system employs standard login credentials for account security, it does not yet include two-factor authentication, an added layer of protection that would enhance user trust and safeguard against unauthorized access

5.2 Future Enhancements

- **Advanced Recommendation Techniques:** Incorporate more sophisticated machine learning models, such as collaborative filtering or hybrid approaches, to provide more accurate and context-aware product suggestions.
- **Integration of User Interaction Data:** Utilize recorded user interactions—such as ratings, browsing duration, and product views—in the recommendation algorithm to deliver highly personalized and adaptive recommendations.
- **Expansion of Sustainability Certification Data:** Partner with recognized certification bodies and integrate official sustainability data to strengthen the credibility of eco-scores and enhance user confidence in the platform.
- **Implementation of Two-Factor Authentication (2FA):** Introduce an additional security layer through 2FA to improve account safety and foster greater trust among users.

5.3 Recommendation

To address the identified limitations and enhance the overall performance of altEco, it is recommended to adopt more advanced machine learning methodologies that move beyond basic similarity measures, thereby enabling more accurate and context-sensitive product recommendations. Incorporating user interaction data, such as browsing patterns and product ratings, into the recommendation algorithm would further support personalized user experiences. Additionally, the integration

of verified sustainability certification data is essential for improving the credibility and transparency of eco-score assessments. Finally, implementing two-factor authentication would strengthen system security and foster greater user trust. Collectively, these enhancements would contribute to a more robust, reliable, and user-centered platform for promoting sustainable consumption.

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APPENDICES

A. Gantt Chart

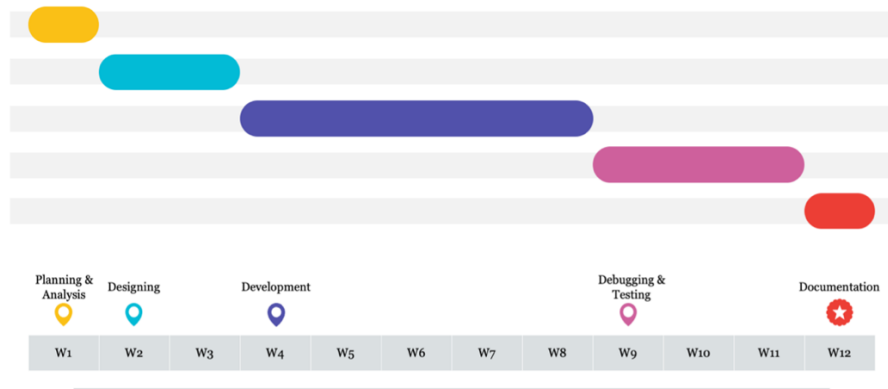


Figure A.1.: Gantt Chart

B. Screenshot

The screenshot shows a web browser window with the URL 'localhost'. The altEco logo is in the top left. The navigation bar includes 'Home', 'About Us', 'EcoScore Calculator', 'Login', and a highlighted 'Signup' button. The main content area is titled 'Create Account' with the subtitle 'Join us today and start your journey'. The form includes fields for 'Full Name', 'Email Address', 'Password' (with a 'Min. 6 characters' hint), 'Confirm Password', 'Date of Birth' (set to '29/07/2025'), and 'Gender' (a dropdown menu). A green 'Create Account' button is at the bottom of the form. Below the button is a link 'Or continue with' followed by a 'Continue with Google' button. At the very bottom, it says 'Already have an account? [Sign in here](#)'.

Figure B.1: Sign-up Page

The screenshot shows the same web browser window as Figure B.1, but the 'Signup' button in the navigation bar is now disabled. The main content area is titled 'Welcome back' with the subtitle 'Sign in to your account'. The form includes fields for 'Email address' and 'Password'. A green 'Sign in' button is at the bottom of the form. Below the button is a link 'Or continue with' followed by a 'Sign in with Google' button. At the very bottom, it says 'Don't have an account? [Sign up here](#)'.

Figure B.2: Login Page

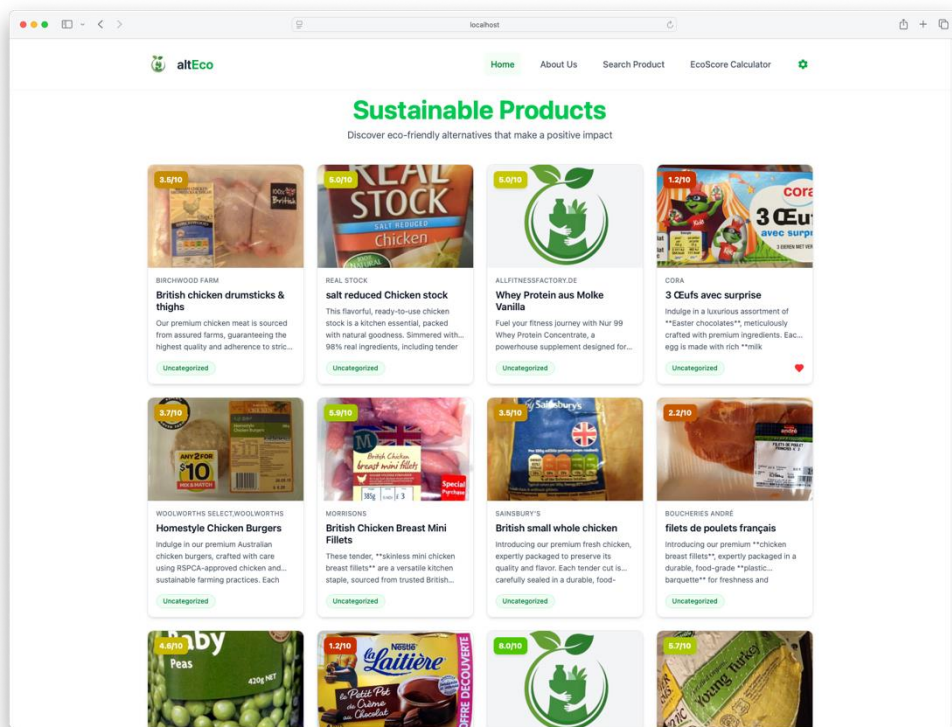


Figure B.3: Home Page

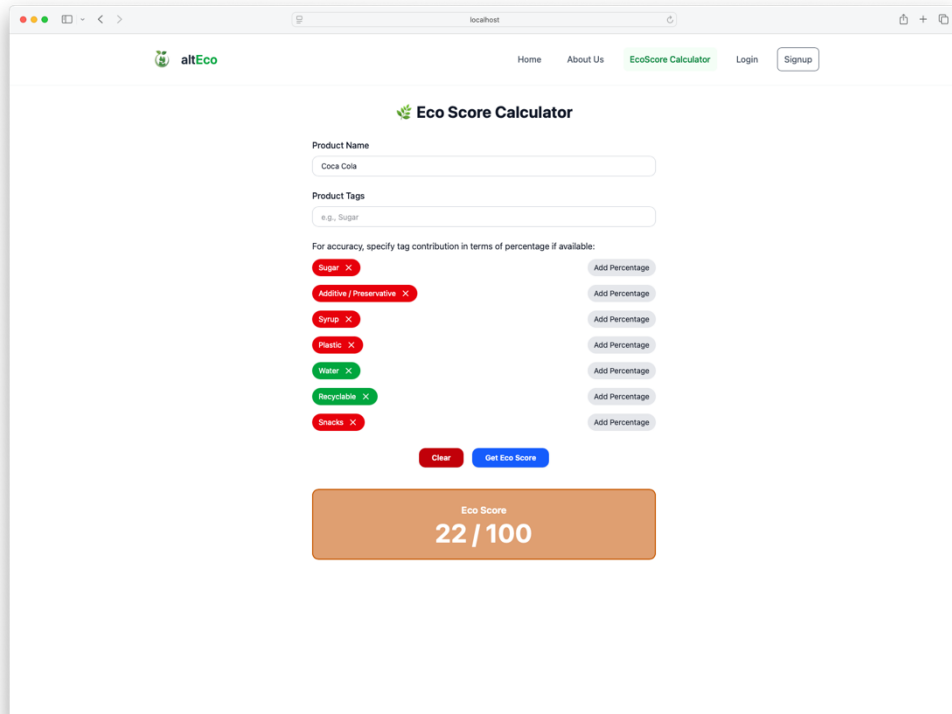


Figure B.4: Eco Score Calculator

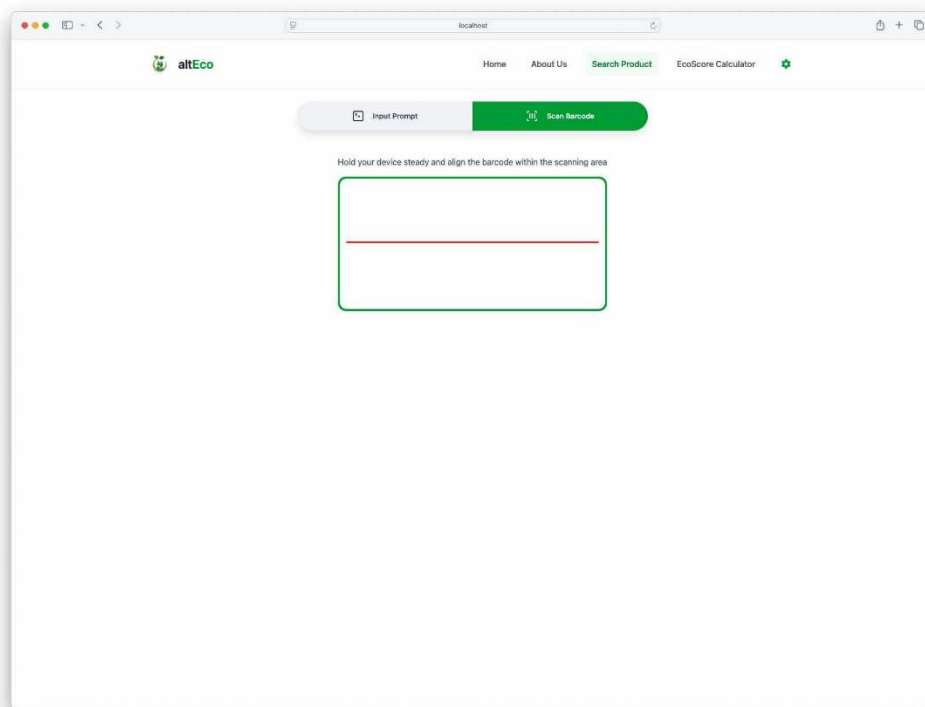


Figure B.5: Barcode Scanner

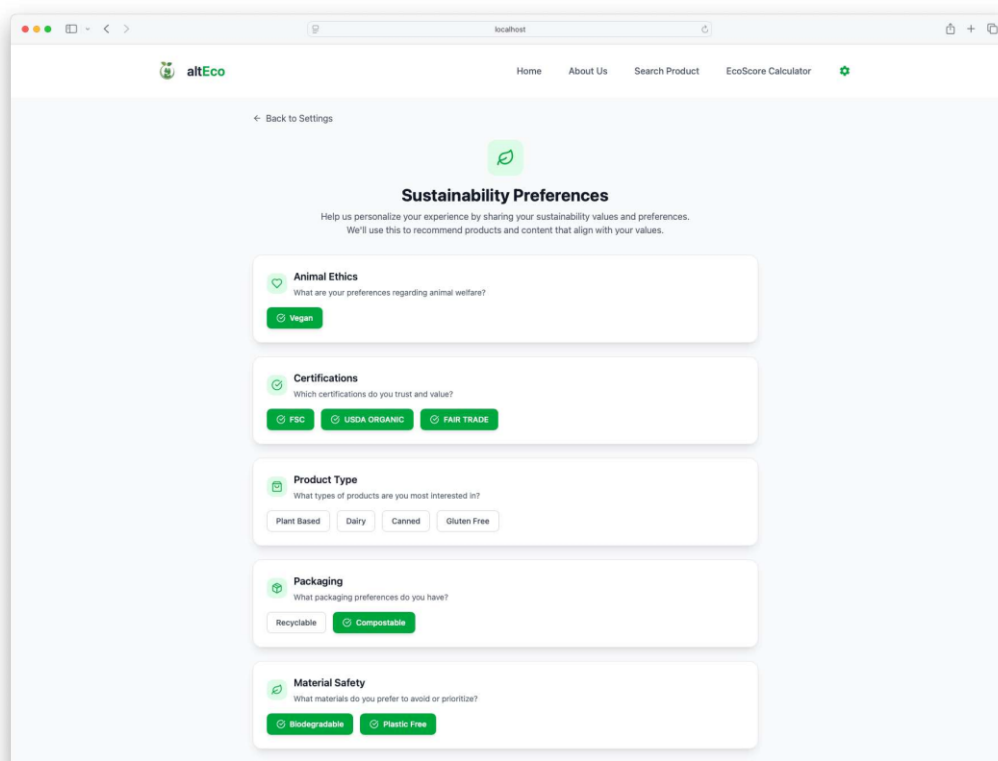


Figure.B.6: Sustainability Preferences

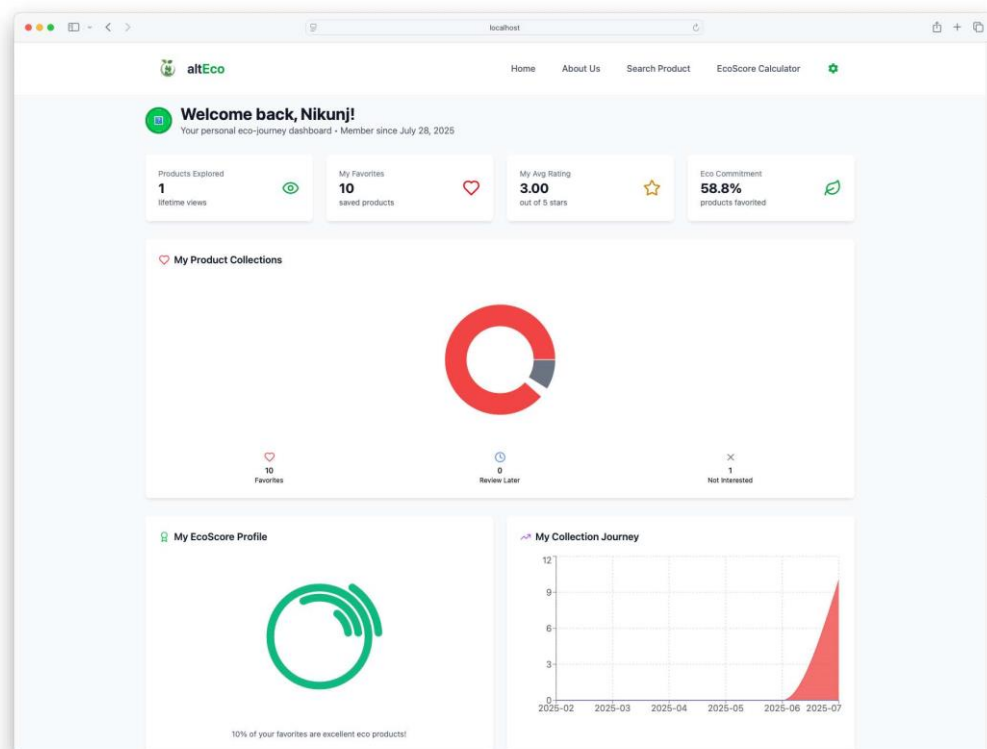


Figure B.7: User Dashboard

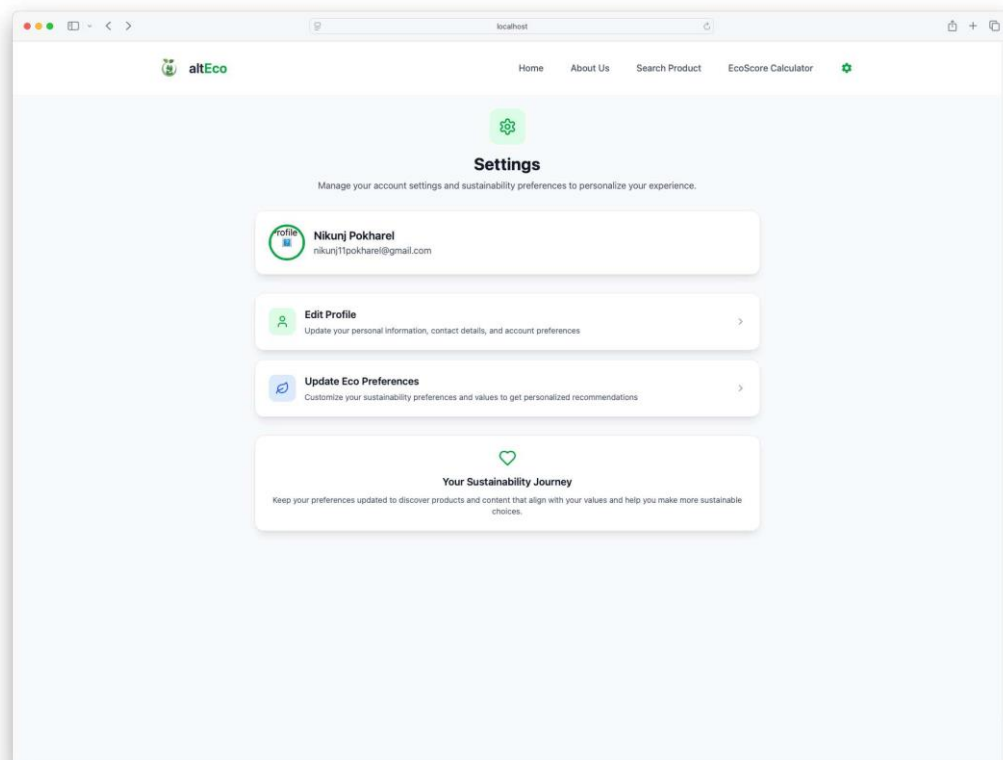


Figure B.8: Settings

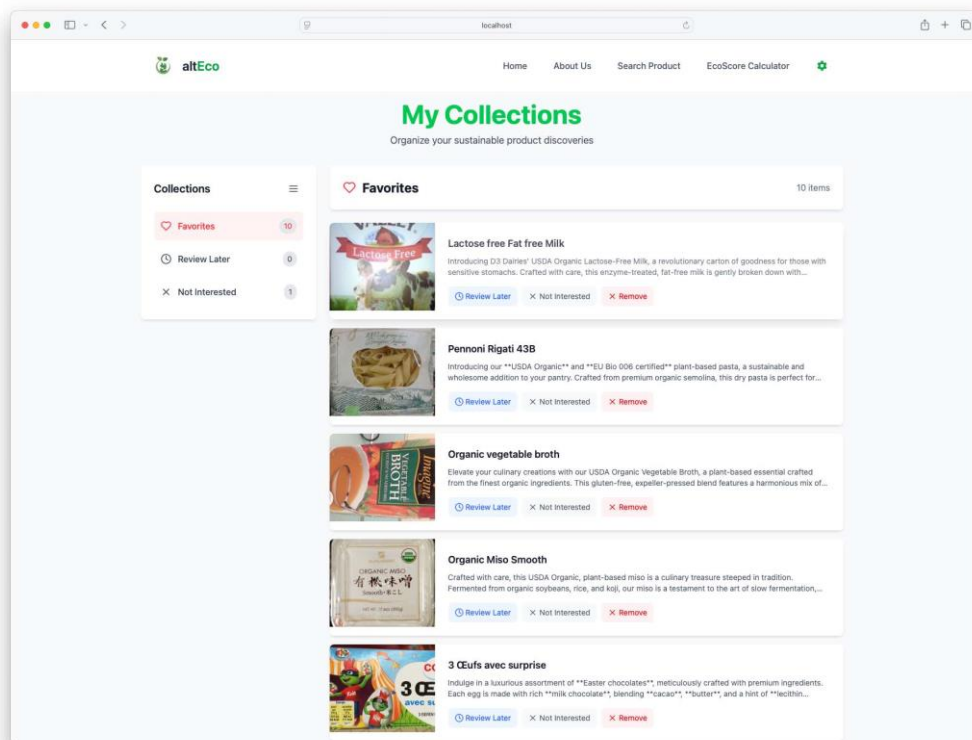


Figure B.9: Collections Page

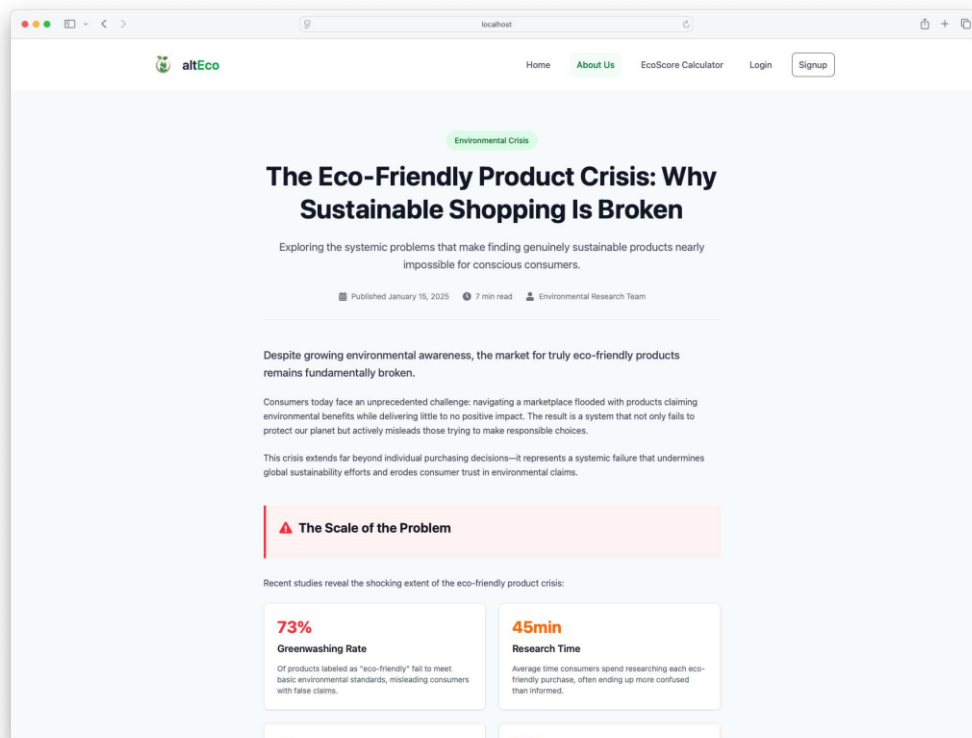


Figure B.10: About Us Page